5. TESTING THE SOLUBILITY OF THE SALT

Aim:

To test the solubility of the given salt based on the saturation and un saturation of the solution at a given temperature.

Principle:

A solution in which no more solute can be dissolved in the solvent at a given temperature is called saturated solution. If the solvent can dissolve more solute than what is present, the solution is called unsaturated solution.

Materials Required:

A 250 ml beaker, a Stirrer, sufficient quantity of distilled water, 100 ml measuring jar, table salt in three packets weighing as 25g, 11g, and 1g.

Procedure:

In a 250ml beaker ,pour 100 ml water using measuring jar. To this water add table salt (25 g) from first packet. stir the content very well. Add the next packet containing 11 g salt followed by constant stirring . Now add the third packet containing 1 g salt . Record your observations.

Observation:

S. No	Amount of salt added	Observation [Salt dissolved/undissolved]	Inference [unsaturated/ saturated/super saturated]
1			
2			
3			

Result: From the above observation, it is inferred that the amount of salt required for saturation is _____ g

6. TESTING THE WATER OF HYDRATION OF SALT

Aim:

To check whether the given sample of salt possesses 'Water of Hydration' or not. To verify the presence of water molecules in the given hydrated salt .

Principle:

Water of crystallization or water of hydration is the phenomenon shown by certain salts in which water molecules are present inside the crystals are responsible for their colour and geometry. e.g. Crystalline copper sulphate $CuSO_4 \cdot 5H_2O$

Materials Required: A pinch of crystalline copper sulphate in a test tube, tongs, spirit lamp.

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